

GM P4 1227165 ECM Simulator

Theory of Operation

CONTROLLED ENGINEERING PUBLICATION

ECM service number 1227165 | Supplied 9340 image
Revision 1.1 | 6 October 2026

Firmware authority: `evidence/firmware/bua-hac.lst`

This publication describes evidence-supported processor-visible behavior and explicitly preserves unresolved custom-device, electrical, mechanical, power, and plant boundaries.

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Document control

Field	Value
System	GM P4 engine control module (ECM), service number 1227165
Firmware basis	Supplied 9340 image and corrected BUA source/listing provenance
Firmware authority	evidence/firmware/bua-hac.lst
Implementation	Strict-C89 behavioral port and deterministic PC simulator
Revision	1.1 - Step 181, 2026-10-06
Status	Controlled engineering publication
Detailed authority	Linked reference chapters and hardware/firmware cross-reference

This document is the master operational description of the repository's evidence-supported system. It explains how the supplied program image, processor-visible hardware, MEMCAL, portable C implementation, and simulator fit together. It consolidates the sixteen completed theory chapters; it does not replace their detailed address, net, calibration, implementation, and regression records.

The behavioral port is closed against the supplied image at the documented processor-visible boundary. That conclusion does not claim cycle-exact processor emulation, complete custom-device reconstruction, electrical equivalence, a production engine model, or safety qualification for connection to physical hardware.

1. Scope and evidence method

1.1 Controlled identity

The hardware identity is ECM service number 1227165. The program basis is the corrected BUA source family associated with the supplied 9340 material. BUA is retained as source and label text; it is not, by itself, proof that every external artifact represents the same calibration, MEMCAL, or board variant.

The authoritative executable record is ``evidence/firmware/bua-hac.lst``. Its emitted instructions, bytes, addresses, targets, tables, and execution order control firmware claims. `bua-hac.txt` is excluded because its corrected-revision status has not been established. Source provenance and image differences are recorded in the [evidence register](#) and ``BUA_SOURCE_PROVENANCE.md``.

1.2 Evidence order and claim status

When evidence conflicts, this work uses the following order:

- 1 verified emitted instructions, addresses, and bytes;
- 2 calibration data together with its executable use;
- 3 repeatable tests, measurements, and independent cross-checks;
- 4 source labels and comments;
- 5 secondary descriptions and historical research.

Claims use the repository statuses Confirmed, Strongly supported, Provisional, Variant-dependent, Unknown, and Boundary. Status belongs to an individual claim, not to an entire artifact. Observable behavior is established before a functional name is assigned. Missing device documentation remains an explicit boundary rather than being completed by analogy.

1.3 Fidelity classes

Class	Meaning in this repository
F0	Listing-backed firmware state, arithmetic, ordering, branch, table, or raw write preserved by the port.
F1	Directly supported hardware connection, schematic path, physical construction, or MEMCAL relationship.
F2	Raw processor-visible input, event, register state, or output observation at the hardware-abstraction-layer (HAL) seam.
F3	Optional simulation assumption: sensor transfer, electrical response, engine/vehicle dynamics, actuator response, or other plant behavior.
F4	External or unresolved behavior: undocumented custom-device internals, unavailable code, physical power/reset effects, or uncharacterized circuitry.

The governing flow is:

Architecture boundary: Physical system / plant (F3-F4) -> raw HAL boundary (F2) -> firmware behavior (F0) -> raw HAL outputs, with schematic-proven connections retained as F1 evidence.

The simulator may add an F3 plant around the F0 core, but an F3 result never upgrades itself into firmware or hardware evidence.

2. System architecture

The ECM combines a processor, volatile and retained state, external program memory, processor-visible custom peripherals, a serial analog-to-digital (A/D) converter, discrete and event inputs, output drivers, and a removable MEMCAL. The supplied program window is \$8000-\$FFFF; the interface exposes A0-A14, D0-D7, /ROMCS, and /ROMOE. Eight vector words occupy \$FFF0-\$FFFE.

The custom-device boundary is central to the architecture:

- U9 exposes the \$3FC0-\$3FFF processor-visible timing/register window. Firmware use establishes particular addresses for reference, injection, spark, dwell, knock, and status. It does not establish U9's internal timer topology or a complete pin-to-register specification.
- U10 (16034988) is the serial A/D boundary. Firmware supplies a selector and receives an 8-bit raw sample through \$F1BE-\$F1DF.
- U11 receives MEMCAL configuration and participates in oscillator, MAP, injector-limit, and reference-related connections. Its internal transfer functions remain F4.
- U12 participates in distributor reference, cylinder/knock selection, ignition/EST feedback, and injector drive/current-control paths. Its internal thresholds, polarity, phase, and control laws remain F4.

The [processor and custom-peripheral chapter](#), [U9 map](#), and [U11/U12 boundary chapter](#) are controlling references. Unclassified locations in \$3FC0-\$3FFF are not assumed to be unused or reserved.

3. MEMCAL and calibration architecture

The MEMCAL is both program/calibration carrier and physical configuration hardware. Its 66-contact relationship to motherboard connector J4, EPROM, two resistor networks, populated jumpers, and CAL paths are part of the system architecture. Original owner-confirmed resistance measurements control replacement resistance targets. Photographs and direct physical evidence control the earlier as-built reconstruction where legible. KiCad, LTspice, and historical notes provide supporting evidence; discrepancies remain visible.

Step 179 records the owner's confirmation that the preserved workbook contains original NetRes pin-to-pin measurements and defines the resistance target. The earlier hand reconstruction is approximate. New passive DC candidates closely match the measured matrices but still require physical validation. Photographs remain authoritative for the earlier as-built population. See the [reconstruction report](#).

Important established anchors include CAL42 to U11 pin 18 OSC, CAL56 to U12 pin 11 CYL, CAL61 to U11 pin 28 MAP, CAL59 to VIGN, and the CAL45/CAL46 common node. These are connection or configuration statements, not licenses to invent the receiving devices' internal behavior. In particular:

- CAL42/U11 OSC is not automatically the U9 timer clock;
- CAL56/U12 CYL is not a PROM byte and must not be equated with LC009 or LC225;
- CAL29's analog ESC route is distinct from the CAL32/U12/U9 knock-event route;
- resistor-network values or topology that remain visually or electrically ambiguous remain unresolved.

The detailed reconstruction is in [MEMCAL architecture](#), [functional networks](#), and [motherboard/firmware paths](#).

3.1 MEMCAL/U11 bias, configuration, and injection block

Schematic location. Sheets 4 (Ignition-Injection) and 5 (Connectors), U11 16054995, U12 16034984, J4, and both NetRes packages. Original package terminal numbering and complete mappings are in the [Step-180 circuit review](#).

Electrical operation. For the preferred measured-matrix candidate, 375 pin 14 is VCC; pins 4 and 8 are ground. The measured 7-8 link grounds U11 pin 8. The pin-5/6/9 resistor group is a coupled bias network, including the 267-k Ω branch between U11 pins 2 and 13. The motherboard adds 51.1 k Ω from VIGN and 5.49 k Ω to ground at the pin-9/U11-13/26 node, so its voltage is not set by the MEMCAL alone. The 88.7-k Ω branch pulls U11 OSC toward VCC; the 0.033- μ F capacitor is on the separately labeled U11 C terminal. An internal oscillator equation is not established.

The 376 network provides VCC pull-ups, a 150-k Ω /1.4-k Ω divider, a VIGN-derived 24.9-k Ω /130-k Ω divider, and a 7.5-k Ω ground return at U12 CYL. Its 8.25-k Ω resistor is on the MEMCAL side of the series capacitor leading to U11 pin 16. The measured 7-9 link routes conditioned TPS at J4-64 to J4-61/U11 pin 28, labeled MAP. Actual conditioned MAP at J4-63 is isolated inside this package. This is a mapping deduction pending original-hardware continuity confirmation; it does not short external MAP and TPS together.

Firmware operation. The NetRes resistors are fixed electrical configuration, not programmable EPROM tables. Firmware consumes U9 reference occurrence and period state (\$CAC6-\$CAD3, \$CB5A-\$CB5D) and derives RPM at \$CDE6-\$CE41. Cylinder/configuration consistency checks at \$F682-\$F68B compare masked \$002F state with LC225; Error 41 qualification includes \$E6A1-\$E6AA. These paths do not establish U12's analog CYL decoding or equate its resistor with the separate normalization constant LC009. Ordinary firmware load remains VMAF-derived, not a conversion of the NetRes's U11 MAP-labeled terminal.

Complete signal path. MEMCAL configuration and motherboard bias act at U11/U12; engine references and sensors also reach these chips. U11 emits INJ/INJLIMP. U12's reference and injector interfaces connect to U9 and the injector output stage. Firmware commands processor-visible timing state; it does not directly calculate the undocumented U11 analog transfer function.

Fault behavior. Incorrect resistors, missing ground links, or open carrier contacts can alter configuration/bias even with a good EPROM. Isolated resistance matches do not prove loaded voltages, oscillator timing or backup fueling. The ~LIMP net crosses U12, output gating, and power circuitry. ALDL mode, firmware sensor substitutions, and hardware backup operation are separate concepts. A MEMCAL-configured backup injection role is strongly supported by these external connections; exact entry conditions and fueling equations remain unknown. No claim is made that every diagnostic error asserts ~LIMP.

Evidence boundaries. Candidate DC accuracy is established numerically against the owner's original measurements; original hidden topology and powered/dynamic behavior are not. Do not infer a frequency from $1/(RC)$, a cylinder count from 7.5 k Ω alone, or an injector pulse width from unloaded divider voltages.

4. Reset, startup, and retained state

Reset vectors enter the source-ordered power-on path at \$C800-\$C9F3. That path clears the specified volatile state, samples socket/checksum and mode conditions, selects factory or normal operation, handles the optional-ROM decision, validates retained state, performs recovery when required, and enters normal initialization. The port preserves this order rather than replacing it with a generic initialization routine.

Retained validation includes the listing-defined checksum computation at \$F3A7-\$F3B4. Recovery at \$F434-\$F446 stores \$8000 in both save-area marker words and initializes all sixteen block-learn-memory (BLM) cells to \$80. Normal startup then establishes controller state, initial sensor samples, engine-running/reference state, injector and spark state, and initial idle-air control (IAC) positioning before periodic operation.

Factory selection executes a separate listing-backed control and test path. The port preserves its RAM fill, table, checksum, A/D scan, communications, raw output exercises, and restart decisions where emitted code is available. The external factory fixture and FMD exchange remain boundaries.

Physical VIGN/VBATT rail timing, keep-alive supply behavior, processor reset electrical consequences, unavailable optional code at \$5800, and vector targets at external \$6000 remain F4. The simulator exposes decisions and events at F2; it does not claim to reproduce those physical mechanisms.

5. Runtime scheduler and event model

Ordinary IRQ service has a 6.25-ms cadence. The source alternates odd and even minor paths, so each minor path executes every 12.5 ms. A one-of-sixteen major dispatcher selects one segment per IRQ; each particular major segment recurs every 100 ms. Common service ordering, including the 6.25-ms IAC motor executor, is preserved.

Execution domain	Established cadence or trigger	Principal role
Common IRQ	6.25 ms	common acquisition/service, IAC one-step execution, shared state
Odd minor	12.5 ms	airflow/load/fuel-oriented work
Even minor	12.5 ms	reference/spark/O2-oriented work
Major segment	one of 16 per IRQ; 100 ms per segment	slower state machines, diagnostics, outputs, accessories
Reference event	external engine event	occurrence/period/counter state for RPM, injection, dwell, and spark

Reference events are not scheduler ticks. A host or target supplies raw engine events/state through the HAL; firmware consumes them in the established source order. Collapsing both domains into one periodic host loop changes observable firmware behavior. The detailed order is documented in [Interrupt and scheduler architecture](#).

6. Input acquisition and signal processing

Firmware explicitly selects U10 channels and receives an 8-bit sample. Normal operation does not perform one continuous universal scan; cadence depends on common, minor, major, startup, or factory execution context.

Selector	Channel/net	Supplied-image role
\$00	AN0 MAP2	factory scan; no explicit normal request found
\$10	AN1 VOLT	voltage/battery-related processing
\$20	AN2 O2	oxygen acquisition and filtering
\$30	AN3 MAP	factory scan; no explicit normal request found
\$40	AN4 CTS	coolant conversion
\$50	AN5 TPS	throttle acquisition, learning, normalization
\$60	AN6 PUMPVOLT	pump-voltage diagnostics
\$70	AN7 DIAG	mode selection

Selector	Channel/net	Supplied-image role
\$80	AN8 MAT	air-temperature processing
\$90	AN9 ESC	factory scan; no explicit normal request found
\$A0	AN10 VMAF	analog mass-airflow path
\$B0	unresolved	factory scan only; channel identity unresolved

Raw counts, learned references, filtered state, and engineering-unit or plant interpretation remain separate. Representative states include TPS raw \$0081 and normalized \$0082, VMAF raw \$00ED and airflow \$00EA:\$00EB, voltage \$007E, pump voltage \$007F, MAT \$012B/\$0060, and current load \$0063.

The supplied image's principal load path is VMAF-derived airflow multiplied by reference period \$0095:\$0096 at \$D769-\$D7A0. Physical MAP/MAP2 resources and the CAL61/U11 terminal labeled MAP exist (the measured 376 jumper routes TPS to that terminal), but the evidence does not make them the ordinary producer of \$0063. Vehicle speed and distributor reference are discrete/event paths, not A/D channels.

7. Reference, RPM, dwell, spark, and knock

Conditioned distributor activity reaches U12 REF; U12 presents reference- related connections to U9/U11, and U9 makes occurrence and timing state processor-visible. Firmware qualifies reference state, derives RPM using the image's supported V8 geometry, and maintains no-reference and engine-running state. Four distributor reference pulses per crankshaft revolution are strongly supported; a reference interval represents 90 crank degrees.

Spark production combines calibrated main and coolant terms, startup behavior, mode terms, limits, hot-restart effects, and knock retard. At \$D20C-\$D267, the firmware converts desired angle to timing relative to reference period and stages U9 state. Important processor-visible locations include \$3FC8 current period state, \$3FDC dwell period, \$3FE4/\$3FE6 next-dwell state, \$3FE8 fire/fall delta, \$3FEC last-reference counter, \$3FF6 reference-to-fire offset, and \$3FFC control/status.

Knock is a separate event path. CAL32/external KNOCK# reaches U12 and U9; firmware consumes changes in \$3FCA at \$D0D1-\$D157, accumulates retard in \$00A5, applies it to spark, and recovers it in Segment A at \$EB3A-\$EB59. Error 43 uses the associated activity checks. CAL29/U10 AN9 remains a distinct analog ESC observation and is not inserted into this event path.

U9/U12 physical edge generation, timer scaling, EST and bypass polarity, ignition-coil current, and exact feedback transformation remain F4. Error 42 establishes firmware feedback-consistency behavior, not the undocumented electrical implementation.

8. Airflow, fuel, and injection

Normal airflow begins with U10 selector \$A0. \$F7AC-\$F7B6 stores VMAF raw state at \$00ED; subsequent processing produces airflow \$00EA:\$00EB. \$D769-\$D7A0 combines airflow with reference period to produce load \$0063, while shifting prior values through \$0061/\$0062.

The fuel path then selects cranking or normal operation and applies the listing-defined commanded air/fuel relationship, startup and power enrichment, oxygen feedback, BLM correction, acceleration enrichment, deceleration fuel cutoff (DFCO), battery-related compensation, pulse-width shaping, and service gates. The sixteen-cell BLM uses a four-by-four airflow/RPM selection for this image (LC017=0) and remains distinct from the faster closed-loop correction.

The established command chain is:

VMAF -> airflow -> load/fuel terms -> corrected pulse width -> permitted injector service -> U9 \$3FD0

U9 \$3FCE is independently exercised as EFI delay. U9 INJS/INJA, U12 injector drive/current control, U11 INJ/INJLIMP, Q1, and the external injector circuit form the hardware envelope. Firmware calculation and raw commands are F0/F2; phase, current limiting, waveform, opening delay, fuel pressure, flow, and combustion are

F3/F4.

9. Idle-air control

Firmware builds a coolant- and mode-dependent target idle, applies minimum- position and anticipation state, and runs the established feedback regulator every 50 ms. The regulator preserves source-specific proportional, derivative, deadband, residual, and fractional-step behavior. It produces a packed direction/magnitude request; a separate 6.25-ms motor service consumes at most one step and updates phase and position bookkeeping.

Throttle follower and A/C, fan, and calibrated auxiliary load anticipation add feed-forward demand without replacing feedback regulation. Startup establishes initial position state. During key-off, \$D370-\$D3DB requests closure to the hard stop, initializes the software counter, then reopens toward calibrated park LC62F=144 through repeated ordinary motor services.

The \$002C HAL observation is software position bookkeeping, not measured pintle position. Winding current, direction at the connector, missed steps, hard-stop mechanics, airflow per step, and engine response remain F3/F4.

10. Emissions, accessories, and transmission

The major loop qualifies secondary-air (AIR), exhaust-gas-recirculation (EGR), canister-purge/CCP, cooling-fan, and air-conditioning requests using calibrated thresholds, delays, and hysteresis. A/C and fan state also feed the IAC anticipation path. Subsystem request state remains distinct from common output staging and from the physical valve, relay, fan, or compressor state.

Transmission logic combines selector/transmission state, vehicle speed, and Segment-E torque-converter-clutch (TCC) qualification. Three stages must remain distinct: firmware TCC request, raw ECM output, and actual powered/hydraulic clutch state. The external brake series-power path is not silently rewritten as an ECM software input. Gear ratios, converter slip, and vehicle dynamics in the optional PC drive model are F3 assumptions; driver, solenoid, and hydraulic behavior not directly established remain F4.

11. Output staging and electrical interfaces

Major Segment 1 \$EDA3-\$EF03 is the common raw output-staging block. It preserves battery qualification, ordinary and field-diagnostic engine-off patterns, normal subsystem selection, Mode-4 pulse-width-modulation overrides, and direct writes to \$3FCC, \$3FD2, \$3FD4, \$3FD6, \$3FD8, and \$4004. The output regression freezes the integrated raw-stage signature FAADF8A6.

A register value is not automatically a voltage, current, relay state, valve position, or actuator force. The supported output sequence is:

firmware request (F0) -> raw processor-visible write (F0) -> proven connection (F1) -> HAL observation (F2) -> optional plant interpretation (F3)

Undocumented driver or custom-device transformation stays F4. This applies to injectors and ignition as well as IAC, TCC, AIR, EGR, purge, fan, A/C, service lamp, and factory-test outputs. Raw values such as 0, 1, \$D000, or \$DFFF do not establish active-high/low, sourcing/sinking, or energized-load semantics by themselves.

12. Diagnostics and communications

Diagnostics separate current qualification, stored history, service-lamp request, and physical lamp behavior. Fault-specific counters and state qualify conditions before history is latched or cleared. Field-service mode sequences flash codes through firmware state; electrical lamp polarity remains at the output boundary.

The normal 160-baud asynchronous link is a pulse-width-coded display stream. The completed raw HAL retains its diagnostic and factory managers, processor- visible timing/state, and output observations without claiming external wire voltage or scan-tool presentation.

The 8192-baud path uses the sole device \$80 serial core initialized at \$C9F4 and implemented through \$FA58-\$FC71. Firmware validates receive state/checksum, constructs Modes 0-4 responses, and advances transmit state. Mode 4 becomes active only through the scheduler-facing lifecycle at \$CB67-\$CB72; entry-only commands and exit cleanup retain their source order.

External ALDL transceiver voltage, polarity, collision behavior, precise physical bit timing, unavailable memory reads, and scan-tool formatting remain F3/F4. Protocol state is not evidence of those electrical properties.

12.1 DIAG mode selection and dashboard data block

Schematic location. Sheet 2 DIAG# input, sheet 1 U10 AN7/pin 8, sheet 3 U2 ALDL transceiver, and sheet 5 DIAG#/ALDL# connector paths. DIAG# and ALDL# are separate electrical nets: one requests mode; the other carries serial data.

Electrical operation. DIAG# has a 10-k Ω pull-up to the logic supply and a 10-k Ω series path to DIAG/U10 AN7, with a capacitor at the converter node. Ignoring converter leakage, an external 10-k Ω resistor to ground makes the connector voltage approximately half the pull-up rail. The input series resistor and capacitor condition the signal; they do not halve its unloaded steady-state voltage again. An open solder joint at U10 pin 8 can disconnect the internal converter input from the correctly biased external trace. Its resulting reading is not predictable without leakage/internal-device information.

Firmware operation. The common A/D routine is \$F1BE-\$F1DF. Segment 3 at \$EA28-\$EA47 requests selector \$70, clears \$0035 bits 4/5, and compares the sample with literal counts 40, 100 and 152. A count below 40 sets bit 4. Counts 100 through 151 set bit 5 unless \$0046 bit 3 is already set. Other counts leave these two bits clear. These are verified count thresholds, not measured voltages. Startup also samples DIAG and has additional factory/exceptional paths.

At \$F8B6-\$F8E8, bits 4/5 select the serial table: ordinary operation uses \$C6FE, whereas diagnostic selection uses \$C70D. The ordinary table points to LC009, \$011A, \$011E, and LC70C; it carries configuration/scaling and fuel/distance-related data rather than the ordinary scan-tool list. Emitted additions into \$011A at \$E4EE-\$E4F3 and \$F733-\$F739 independently support its fuel accumulation role. Table comments alone do not define engineering units. The ordinary record is commonly observed as five bytes including its mode byte; the diagnostic record is the longer 25-byte stream.

Complete signal path. DIAG voltage -> U10 sample -> mode flags -> serial payload selection and control-mode consumers -> U2/ALDL line -> dashboard or scanner. The dashboard needs the normal economy-related record. Replacing it with the diagnostic record can remove or corrupt the expected economy updates. Range can consequently be affected through the dashboard's economy estimate; the dashboard's precise parsing and range algorithm are not in this ECM image.

Fault behavior. A false ALDL request can cause high idle alongside an invalid MPG display. \$D448-\$D450 tests bit 5 and substitutes \$50 (80) into the RPM/25 target path; this is 2000 RPM in that particular command path, not proof that a faulty engine must physically idle at that speed. Other gates, IAC state, and engine airflow determine the resulting speed. Do not describe the observed 1000-RPM symptom as an unconditional calibrated 1000-RPM ALDL target.

Reported repair example. In the owner's June 2023 [CorvetteForum repair thread](#), posts under dldavis describe roughly 1000-RPM idle, MPG stuck at 1.2, and an unrequested change from the short record to the diagnostic record. The June 24 post reports resoldering A/D pin 8 (DIAG); the June 28 follow-up reports three days of correct operation. The owner confirmed the recollection in this session. This is firsthand repair history, not a reproduced electrical test. A disconnected pin causing erroneous mode samples is an explanation consistent with the repair; its actual disconnected voltage was not measured here. This episode is not proof of hardware limp-mode activation or an incorrect NetRes.

Evidence boundaries. Instructions establish mode selection and payload switching. Exact line voltages, dashboard error handling, and the failed joint's internal electrical behavior remain hardware or external-document boundaries.

13. Key-off, shutdown, and exceptional modes

The live lifecycle composes reset-vector power-on, ordinary IRQ operation, the \$D6D1-\$D769 ignition/key-off transition, retained-state work, IAC homing, and the terminal \$D6EA software-interrupt (SWI) powerdown request. BLM commit via LF447 occurs in its documented order. If ignition returns while the software still executes, restart follows the established path rather than a host-created shortcut.

The vectors at \$FFF0-\$FFFE target \$6000, \$C9F4, \$F27B, \$6000, and four copies of \$C800; \$F27B is an immediate return-from-interrupt. What the processor, power supply, keep-alive circuitry, or external \$6000 target does after SWI/reset is outside the supplied image. The HAL reports the software request; it does not simulate physical loss of power as an F0 fact.

14. C port and embedded boundary

The implementation is intentionally a single strict-C89 translation unit: `main.c` includes the fragments under `src/`, `simulation/`, and `tests/`. Exact-width assumptions are enforced with compile-time checks, and explicitly encoded byte/word operations preserve big-endian processor-visible behavior. The code fragments are not intended to compile independently.

`src/hal_interface.inc.h` is the explicit raw seam. It supplies A/D, reference, discrete, communications, power/startup, and factory stimuli and observes U9, parallel-I/O, injector, IAC, and common output state. Target-specific timers, vectors, A/D and capture peripherals, serial transceivers, output drivers, retained memory, watchdog, and physical power behavior belong behind that seam.

An embedded port must retain F0 order and widths while replacing F2 mechanics. It must not migrate the PC's F3 engine/vehicle plant into firmware or fill F4 U9/U11/U12 behavior for convenience. The controlled sequence and exit criteria are in the [embedded and HIL roadmap](#).

15. Simulation and verification

Verification has three complementary layers:

- focused regressions preserve individual arithmetic, state machines, tables, and raw interfaces;
- scheduler and lifecycle regressions prove source-order composition;
- deterministic PC scenarios exercise closed-loop interaction with explicitly assumed F3 engine and transmission plants.

`make test` compiles only `main.c` using `gcc -std=c89 -Wall -Wextra -pedantic`, runs the complete regression program, checks every Makefile-gated result and signature, and rejects genuine FAIL output. Frozen signatures include the normal-operation signature 4BA6B7C6, transmission-aware signature 9732D09B, Segment-D diagnostic signature F357A5F2, Segment-1 output signature FAADF8A6, and ignition-lifecycle signature 16D17C9C. Later raw-HAL and lifecycle signatures are documented by the verification chapter and audits.

Signatures are broad deterministic change detectors. They do not replace explicit assertions, prove untested paths, or establish physical truth. A change that preserves a signature can still be wrong; a deliberate behavior change requires source evidence, focused checks, integration checks, and a reviewed baseline update.

16. Explicit unresolved boundaries

The following are intentionally unresolved or external to the supplied image:

- undocumented internals, transfer functions, phase, polarity, thresholds, timer scaling, and current-control behavior of U9, U11, and U12;
- unclassified U9 \$3FC0-\$3FFF locations and unproven pin/register mappings;
- U10 selector \$B0, analog reference/accuracy details, and variant-specific normal use of MAP2, MAP, or ESC channels;

- exact physical relationships among OSC, CYL, distributor reference, ignition, EST/bypass feedback, and knock events beyond established paths;
- injector and ignition waveforms, driver protection, coil/injector current, output polarity, and load electrical behavior;
- IAC mechanics and airflow, relay/solenoid/valve/fan/compressor response, TCC hydraulics, sensor transfer functions, and engine/vehicle dynamics;
- optional code at \$5800, external vector targets at \$6000, the factory fixture/FMD implementation, and unavailable external memory contents;
- physical reset, rail timing, standby/keep-alive power, watchdog, and processor consequences after SWI or power transition;
- variant applicability of external photographs, MEMCAL networks, calibrations, and historical artifacts not directly tied to this unit.

These are engineering work items, not gaps to be closed by plausible guesses. New evidence should update the evidence register and cross-reference first, then the affected detailed chapter and this synthesis.

Appendix A. Consolidated address and register index

This index collects the principal processor-visible locations used in the system narrative. Functional names describe established executable use; they do not imply undocumented electrical or custom-device behavior.

A.1 RAM and software state

Address/range	Established role	Detailed reference
\$002C	IAC software position bookkeeping	Idle-air control
\$005B-\$005F	coolant conversion and diagnostic state	Sensor acquisition
\$0061-\$0063	prior and current load state	Airflow/fuel/injection
\$0064	common temporary A/D result	A/D acquisition
\$006F/\$0071/\$0073	oxygen acquisition/filter state	Sensor acquisition
\$007E	persistent VOLT sample	Sensor acquisition
\$007F	persistent PUMPVOLT sample	Sensor acquisition
\$0081/\$0082	raw and normalized TPS state	Sensor acquisition
\$0086/\$0087	learned closed-throttle state	Sensor acquisition
\$0095:\$0096	reference-period state used by load/fuel processing	Airflow/fuel/injection
\$00A5	accumulated knock-retard state	ESC/knock chain
\$00DD-\$00DE	transient TPS state	Sensor acquisition
\$00EA:\$00EB	processed airflow word	Airflow/fuel/injection
\$00ED/\$00EF	raw and intermediate VMAF state	Airflow/fuel/injection
\$012B/\$0060	complemented raw and processed MAT state	Sensor acquisition
\$0152/\$0153/\$0156/\$0157	Mode-4 raw PWM override state	Output staging
\$017B-\$0186	factory-test sequential A/D snapshot	A/D acquisition

A.2 Processor-visible peripheral locations

Address/range	Established role	Detailed reference
\$3FC8	current spark/reference-period state	U9 register map
\$3FCA	knock-event quantity consumed by firmware	ESC/knock chain
\$3FCC	Segment-1 raw MPU output word	Output staging
\$3FCE	EFI delay state	Ignition/injection chain
\$3FD0	synchronous injector command	Airflow/fuel/injection
\$3FD2/\$3FD4/\$3FD6/\$3FD8	Segment-1 raw MPU output words	Output staging
\$3FDC	dwelling-period value	Reference/RPM/dwelling/spark
\$3FE4/\$3FE6	next-dwelling timing and update state	Reference/RPM/dwelling/spark
\$3FE8	current fire/fall delta	Reference/RPM/dwelling/spark
\$3FEC	counter value at last reference	Reference/RPM/dwelling/spark
\$3FF6	reference-to-fire offset	Reference/RPM/dwelling/spark
\$3FFC	control/status word used by EST/bypass and other modes	U9 register map
\$4004	parallel-I/O byte modified by Segment 1	Output staging

A.3 Executable regions and external boundaries

Address/range	Established role	Detailed reference
\$5800	optional-ROM boundary; code unavailable	Startup/shutdown
\$6000	external vector destination; behavior unavailable	Startup/shutdown
\$8000-\$FFFF	supplied program window	Processor/memory architecture
\$C800-\$C9F3	reset and power-on sequence	Startup/shutdown
\$C9F4, \$FA58-\$FC71	device \$80 SCI core	Diagnostics/ALDL
\$CB67-\$CB72	scheduler-facing Mode-4 activation	Diagnostics/ALDL
\$D0D1-\$D157	knock-event consumption and retard	ESC/knock chain
\$D20C-\$D267	spark angle-to-time staging	Reference/RPM/dwelling/spark
\$D370-\$D3DB	IAC reset/homing state machine	Idle-air control
\$D6D1-\$D769	key-off transition and adjacent lifecycle path	Startup/shutdown
\$D769-\$D7A0	airflow/reference-period load producer	Airflow/fuel/injection
\$D6EA	terminal SWI software-powerdown request	Startup/shutdown
\$EDA3-\$EF03	Segment-1 raw output staging	Output staging
\$F1BE-\$F1DF	common U10 A/D transaction	A/D acquisition
\$F3A7-\$F3B4	retained checksum computation	Startup/shutdown
\$F434-\$F446	retained-state recovery initialization	Startup/shutdown
\$FFF0-\$FFFE	emitted vector table	Startup/shutdown

The complete function-to-listing, RAM/register, calibration, schematic, C, and test mapping remains in the [hardware/firmware cross-reference](#).

Appendix B. Detailed chapter map

Chapter	Controlling reference
1	System overview and evidence method
2	Power, reset, retained power, startup, and shutdown
3	Processor, memory, and custom peripherals
4	MEMCAL architecture and reconstruction
5	Interrupt and scheduler architecture
6	Reference, RPM, dwell, and spark
7	Sensor acquisition and filtering
8	Airflow, fuel, and injection
9	Idle-air control
10	Emissions and accessory control
11	Transmission and TCC
12	Diagnostics and ALDL
13	Output staging and electrical interfaces
14	Startup, shutdown, and exceptional modes
15	C port and embedded migration
16	Simulation and verification

The [theory-of-operation index](#) retains the acceptance criteria and supporting signal-chain references for the sixteen-chapter source set.

Appendix C. Glossary

Term	Meaning
A/C	Air conditioning.
A/D	Analog-to-digital conversion.
AE	Acceleration enrichment.
AFR	Air/fuel ratio.
AIR	Secondary-air management system.
ALDL	Assembly Line Diagnostic Link.
BLM	Block-learn memory; retained adaptive fuel-correction state.
CCP	Canister-control purge; used here for the canister-purge function.
CTS	Coolant temperature sensor.
DFCO	Deceleration fuel cutoff.
ECM	Engine control module.
EGR	Exhaust-gas recirculation.
ESC	Electronic spark control; the analog observation path is distinct from the knock-event path.
EST	Electronic spark timing.
FMD	Factory-mode data/exchange boundary used by the factory-test path.
HAL	Hardware abstraction layer; the raw processor-visible F2 seam.
HIL	Hardware-in-the-loop verification.
IAC	Idle-air control.
IRQ	Interrupt request; ordinary periodic firmware service in this document.
MAF	Mass airflow.
MAP	Manifold absolute pressure.
MAT	Manifold air temperature.
MEMCAL	Removable memory/calibration and configuration carrier.
MPU	The custom processor-visible peripheral/register resource named by the source material.
PROM	Programmable read-only memory containing program and calibration data.
PWM	Pulse-width modulation.
RPM	Revolutions per minute.
SCI	Serial communications interface used by the 8192-baud path.
SWI	Software interrupt; used by the terminal powerdown request path.
TCC	Torque-converter clutch.
TPS	Throttle position sensor.
VIGN	Ignition-switched supply/sense domain.
VMAF	Analog mass-airflow voltage net presented to U10.
VSS	Vehicle-speed signal.

Appendix D. Schematic functional-block coverage

The theory retains subsystem chapters, with an electrical-to-firmware block structure: schematic location, electrical operation, firmware operation, complete signal path, fault behavior, and evidence boundaries. Sections 3.1 and 12.1 are the first expanded treatments. The index below covers all six schematic sheets and identifies where further component-level expansion is still needed. A row is a coverage entry, not a declaration that the custom chip is fully explained.

Sheet / block	Electrical boundary	Firmware relationship / chapter	Coverage state
1: U8 processor, crystal, memory bus	Clock, reset, address/data and J4 EPROM interface	Startup and execution; 2, 4, 5	Existing overview; expand physical timing
1: U9 timing/output peripheral	Reference, MAF, VSS, timing and output pins	Register window and event scheduling; 5, 7, 8, 11	Firmware established; internal circuitry bounded
1: U10 A/D and SPI	Analog channels, reference rails, MOSI/MISO/SCK/select	Sensor acquisition; 6	Channel map established; electrical expansion pending
1: U12 SPI/discrete I/O	Digital inputs, CTS range output, reset and LIMP	Input/output state and exceptional modes; 4, 6, 11, 13	Pin roles mapped; internal logic bounded
2: discrete input conditioning	Pull-ups/downs, series resistors and capacitors	Switch/accessory states; 10	Existing subsystem coverage; expand each input
2: VOLT and PUMPVOLT	33.2-kΩ/8.06-kΩ dividers and filtering	Voltage state/diagnostics; 6, 12	Existing acquisition; loaded behavior bounded
2: MAP/MAP2 and TPS	Series filtering and bias paths	TPS learning; MAP factory channels; 6	Existing acquisition plus 3.1 TPS routing correction
2: CTS and MAT	Bias, filtering and CTSHI range circuit	Temperature conversion; 6, 9	Existing firmware; expand range-switch circuit
2: O2 and U4	Differential sensor amplifier and reference supply	O2 filtering/closed-loop fuel; 6, 8	Existing firmware; amplifier transfer bounded
2: ESC and KNOCK	Separate analog ESC and U12 digital KNOCK paths	Factory ESC versus U9 knock counter; 6, 7	Separation established; filter internals bounded
2: VSS and FMAF	U12 signal conditioning with input bias/filtering	Speed/events and MAF interfaces; 6, 8, 10	Existing firmware; conversion details bounded
2/1: DIAG	Pull-up, series resistor, AN7	Mode decoding and data-table selection; 12.1	Expanded with firsthand repair example
3: U13 pump driver	PUMP command to PUMP#	Pump request; 10, 11	Existing output boundary; driver expansion pending
3: U7 IAC driver	IACA/IACB/IACEN to four winding connections	Step sequencing and regulator; 9, 11	Existing firmware; winding/load behavior bounded
3: U3/U6 quad drivers	Output protection, fault feedback and limp gating	Accessory outputs and SES; 10, 11	Existing staging; expand gate/polarity mapping
3: U5 fan/limp gates and RC branch	LIMP-conditioned fan/output logic and delay network	Output and exceptional-mode boundaries; 11, 13	Circuit visible; complete truth table pending
3: U2 ALDL transceiver	TX/RX/enable and shared ALDL# line	160-baud and 8192-baud protocols; 12	12.1 payload path; physical transceiver bounded
4: distributor/reference and EST	Differential conditioning, REF, EST and BYPASS feedback	Reference, RPM, spark and Error 42; 7	Existing firmware; comparator/feedback expansion pending
4: U11/MEMCAL configuration	Fixed bias, OSC, sensors and INJLIMP	Auxiliary injection/custom-chip boundary; 3.1, 8	Expanded; analog equations unknown
4: U12/Q1 injector output	Gate drive, protection, sense resistor, INJLOOP	Injector timing and feedback; 8, 11	Existing firmware; loaded driver behavior bounded
5: J1/J2 vehicle and J3/J4 interfaces	Physical routing, supplies, programming and MEMCAL	Cross-reference for all chapters	Routing index; orientation check still physical

Sheet / block	Electrical boundary	Firmware relationship / chapter	Coverage state
6: U1 power/reset and retained rails	VBATT/VIGN, VCC, standby, reset and LIMP connections	Reset, retention, key-off; 4, 13	Existing firmware; internal power sequencing bounded

Detailed chapter destinations and register anchors remain in Appendix B and the [hardware/firmware cross-reference](#). Next expansions should cover power/reset/LIMP activation, the injector driver, and the remaining analog conditioning blocks using this same structure.